BTS-I(S)-06-24-2496 Reg. No. | | | | | | | | |

C

B.Tech. Degree I Semester Supplementary Examination June 2024

CE/ME/SE 23-200-0102 A ENGINEERING CHEMISTRY

(2023 *Scheme*)

Time: 3 Hours Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Explain the basic concepts of chemical thermodynamics and quantum chemistry.
CO2: Illustrate the spectroscopic methods in characterizing materials.
CO3: Develop electrochemical methods to protect different metals from corrosion.
CO4: Interpret the chemistry of a few important engineering materials and their industrial applications.
CO5. Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(2 \times 5 = 10)$	Marks	BL	CO	PO
I.	(a)	Explain the first law of thermodynamics with mathematical expression.	2	L1	1	3
	(b)	Write down all the quantum numbers for (i) valence electron of Potassium (Z = 19) (ii) valence orbit of Neon (Z=10).	2	L2	1	2
	(c)	Write the principle behind microwave spectroscopy.	2	L1	2	1
	(d)	Explain the significance of Nernst equation.	2	L1	3	1
	(e)	Write a note on the salient features of refractories.	2	L1	4	3

PART B

		$(4 \times 10 = 40)$				
II.		Explain Pattinson's process with a neat labelled phase diagram.	10	L2	1	2
		OR				
III.	(a)	Derive the relation $\Delta G^{\circ} = RT \ln K_p$.	7	L2	5	3
	(b)	Obtain the equilibrium constant K_{eq} for the following reaction at 400 K. $2NOCl \rightleftharpoons 2NO(g) + Cl_2(g)$ Given that $\Delta H^{\circ} = 77.2$ KJ mol^{-1} and $\Delta S^{\circ} = 122$ JK^{-1} mol^{-1}.	3	L2	1	2
IV.		Derive the operator form of Schrodinger equation from classical wave equation.	10	L1	1	3
		OR				
V.	(a)	Obtain the forms of hydrogen atom wave functions.	6	L2	1	3
	(b)	For the wavefunction $\psi = A\cos x \; for \; 0 < x < \pi/2$. Find the value of A.	4	L3	1	3

(P.T.O)

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			Marks	BL	CO	PO
VI.	(a)	Discuss the possible electronic transitions, their regions of absorption and shifts in the position of absorption in electronic spectroscopy.	7	L2	2	1
	(b)	TMS is used as internal reference in NMR. Why?	3	L3	2	3
		OR				
VII.	(a)	Draw the low and high resolution ^1H NMR spectra of (i) CH_3 -CH_2 -CH-Cl_2 (ii) CH_3 -CCl_2 -CH_3 (iii) C_6H_5 -CH_2 -CH_3	6	L3	3	1
	(b)	The fundamental vibrational frequency of HI^{127} molecule is $6 6 \times 10^{-13}$ s^{-1} and the internuclear distance is 1.5×10^{-8} m. Calculate the relative force constant.	4	L3	2	3
VIII.	(a)	Write a note on the effects of corrosion and methods to prevent corrosion.	5	L2	3	1
	(b)	Explain the different physical properties to benchmark Lubricants.	5	L2	4	3
		OR				
IX.	(a)	Describe the mechanism of lead storage battery.	5	L2	3	1
	(b)	Write a note on conducting polymers.	5	L2	4	3

Bloom's Taxonomy Levels
L1-22.7%, L2-59%, L3-18.1%.

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